What Is a Taxi-Out Time? Matching Windows, Silent Substitutions and a Generalist Model for the PRC Data Challenge 2026

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Abstract

The PRC Data Challenge 2026 asks participants to predict taxi-out time for departures at ten major European airports. Its final phase is a blind ranking on four months of 2026, which rewards models that generalise rather than models tuned on a public leaderboard. We describe the generalist model we submitted and the analysis of the target that shaped it. The target is not always a taxi time: about 4.8 % of departures have their off-block silently replaced by the scheduled time, and a rarer family carries a date-rollover defect. We show that the airport off-block always lies within 3606 s of the last filed off-block time, on all 2,062,251 matched departures of 2025. This matching window bounds every prediction and makes a schedule substitution impossible on a known subset of flights; together these two consequences remove 8.3 s of error from an earlier model. The final model is a two-regime mixture whose classifier, regressors, mixture exponent and blend of CatBoost and per-airport XGBoost are chosen on a selection fold and scored once on a separate control fold, without leaderboard feedback. On January and July 2025 it reaches 348.3 s of root mean squared error, against 356.6 s for our earlier, partly leaderboard-tuned pipeline. A decomposition of the remaining error shows that 40 % of it sits in 26 aberrant labels with no identifying signal and 40 % in measurement noise on ordinary flights, which no model restricted to the provided columns can remove. We report the negative results that support this reading, including a recalibration on EUROCONTROL's published monthly taxi-out times and a search for co-occurring defects between arrivals and departures.

1 Introduction

Taxi-out time, the interval between off-block and takeoff, is a standard efficiency indicator in air traffic management. It drives fuel burn and associated emissions on the ground, it is a core metric of Airport Collaborative Decision Making, and its predictability conditions departure metering schemes. The literature on its prediction is mature: reported root mean squared errors fall in the range of 75 s to 90 s on carefully measured physical taxiing [2], with queue size at pushback consistently identified as the dominant predictor since [1].

The PRC Data Challenge 2026, organised by the EUROCONTROL Performance Review Unit together with the OpenSky Network, provides twelve months of 2025 movement data at ten European airports. The initial phase ranked predictions for January and July 2026 on a public leaderboard. Shortly before the end, the organisers added a final phase: a single blind submission on a new file covering January, February, June and July 2026 (670,790 departures), with no leaderboard, together with a review of code and documentation. The stated reason was that signs of optimisation against the public ranking had become too strong.

This change of rules is the frame of the present paper. Our public score with the second-campaign pipeline was 319.21 s, in the lower half of a leaderboard then led at 241.75 s, and

several of its settings had been selected or confirmed with leaderboard feedback. For the final phase we rebuilt the model from scratch under one constraint: every setting is chosen on 2025 data, with a selection fold and a separate control fold, and every component rests either on a structural property of the data or on that out-of-time validation.

We organise the paper around five claims.

1. The target is not always a taxi time, and its two corrupted families behave completely differently with respect to prediction: one is recomputable from provided columns, the other has no identifying signal in any field or signature we tested (Section 3).

2. The airport off-block never leaves a ± 3606 s window around the last filed off-block time. This structural property bounds every prediction and rules out schedule substitution on a known subset of flights (Section 4).

3. A generalist model selected without leaderboard feedback outperforms, on the same hold-out, our earlier pipeline that was partly tuned on the leaderboard, and the gain of each selected component transfers from the selection fold to the control fold (Sections 5 and 6).

4. Most of the remaining error is not reducible with the provided columns: it sits in a few aberrant labels and in the disagreement between two off-block definitions (Section 7).

5. Congestion features are screened off by the observed Network Manager off-block, but only in the causal direction, which reflects the difference between forecasting and retrospective reconstruction (Section 8).

All code, the Kaggle notebook that produced the final submission, the complete experiment log with both positive and negative results, and the diagnostics that support every number in this paper are available at `https://github.com/ugo-roccamatisi/prc-data-challenge-2026`.

2 Data and task

The challenge provides 4,167,797 movements, arrivals and departures, at EDDF, EDDM, EGLL, EHAM, LEBL, LEMD, LFPG, LIRF, LSZH and LTFM for the whole of 2025. Each movement record carries the reporting airport's own timestamps: the movement time (takeoff for a departure, landing for an arrival), the block time (off-block or in-block), the scheduled time, and the stand and runway used. Movement records are left-joined onto the Network Manager flight list, which contributes the flight plan and flown trajectory timestamps, in particular the last filed off-block time (LOBT), the estimated off-block time (EOBT) and the actual off-block time of the flown trajectory (AOBT).

In the ranking files, only two fields are blanked, and only for departures: the block time and the taxi time. Everything else remains visible, including the takeoff time, the scheduled time, the stand, the runway, every Network Manager field, and all arrival records with their in-block times and taxi-in durations. The organisers confirmed that the Network Manager columns are admissible inputs in the final phase.

About 1.5 % of departures are never matched to a Network Manager record, in which case every flight-side column is null. For matched flights, the airport off-block and the Network Manager off-block do not measure the same event. The naive estimator $\widehat{y} = \mathrm{MVT} - \mathrm{AOBT}$ gives a root mean squared error of 380.8 s over the whole of 2025, with a per-airport hourly median bias ranging from -118 s at LIRF to 296 s at LTFM (Figure 1). This is not noise around a common definition; it is two definitions. Our models therefore predict the residual of the target over that naive estimator rather than the target itself.

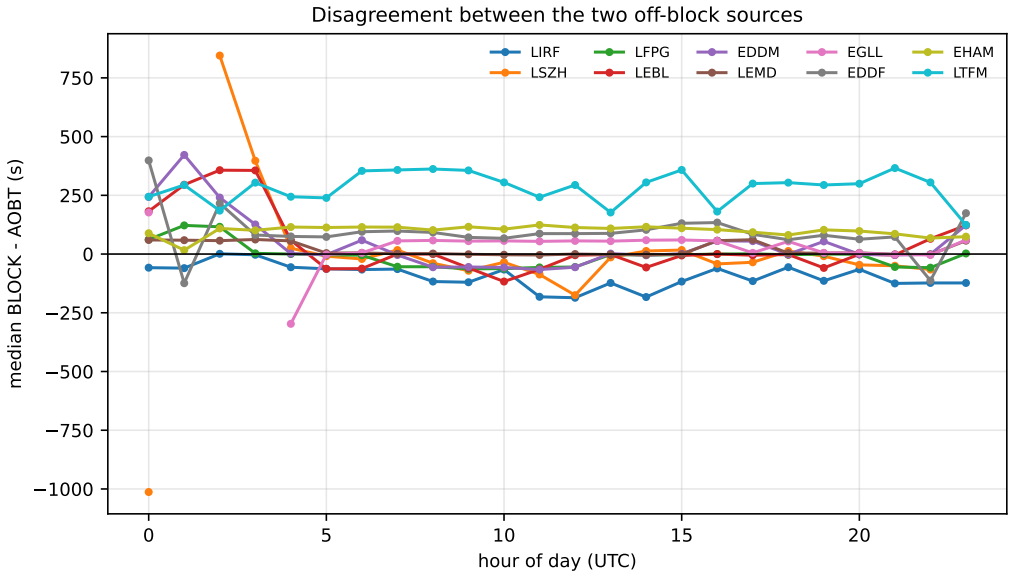


Figure 1: Hourly median of the difference between the airport block time and the Network Manager actual off-block, per airport, 2025. The curves do not scatter around a common zero: the two sources measure different events, and the per-airport offset reaches several minutes.

3 Anatomy of the target

3.1 Schedule substitution

Selecting departures for which the recorded block time equals the scheduled time to within 10 s isolates 4.80 % of departures. On those rows the recorded taxi time is, to within a median of 3 s, exactly MVT − SCHED: the reported quantity is the sum of the departure delay and the real taxi, not the taxi. The median recorded value is 1009 s against 910 s for the rest of the population, so the contamination is invisible in a summary statistic (Figure 3), but these rows carry 29 % of the total squared error of a naive regression. The organisers document that they did not reconcile inconsistencies between the movement and flight sources, which is consistent with what we observe: the substitution is an artefact of the reporting chain, left in place by design.

3.2 A recomputable subfamily

Among the substituted rows, the extreme tail is the one that matters for a squared-error metric. In 2025 there are 76 departures with a recorded taxi time above 20,000 s. Of these, 58 satisfy $|y - (\text{MVT} - \text{SCHED})| < 120$ s, and 72 are at LIRF. On those rows $y - (\text{MVT} - \text{SCHED})$ has a median of 1 s and an interquartile range of 6 s.

Since both the takeoff time and the scheduled time remain visible in the ranking file, these labels are recomputable rather than predictable. The difficulty is identification. A threshold on MVT − SCHED alone is a poor discriminator: at 20,000 s it selects 245 candidates for 57 true positives. The discriminating conjunction involves the join structure rather than the timestamps: requiring in addition that the movement be unmatched to any Network Manager record and that the aircraft type field be empty raises precision from 0.23 to 0.85.

In the second campaign we applied this rule as a hard substitution restricted to LIRF, worth 7.95 s on the leaderboard. The final model keeps the identification but not the airport restriction or the hard decision: for unmatched movements without aircraft type whose MVT − SCHED exceeds 3 h, the label equals MVT − SCHED on 67 % of the 244 such departures of 2025, and

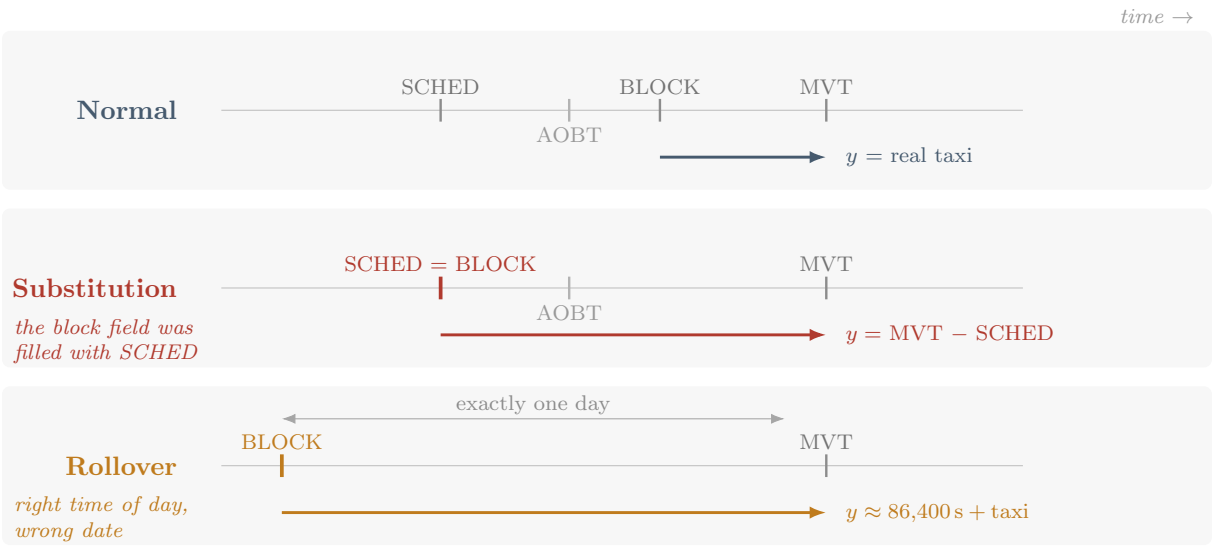


Figure 2: The three regimes of the target, drawn on a common time axis with the takeoff aligned. In a normal record the reported taxi time spans the airport block time to takeoff. Under schedule substitution the block field carries the scheduled time, so the reported value is the departure delay plus the taxi, and is exactly recomputable from two columns that remain visible at evaluation time. Under date rollover the block field carries the correct time of day with the previous day's date, which inflates the value by one day and leaves no visible trace.

the prediction uses that rate, estimated per airport and shrunk towards the global rate, as a mixture weight (Section 5).

3.3 A family without a usable identifier

The remaining 18 aberrant labels of 2025 follow a different mechanism. Fourteen of them lie between 87,168 s and 88,392 s, that is 86,400 s plus an ordinary taxi: the recorded block time carries the correct time of day and the previous day's date. The quantity $y - 86{,}400\,\text{s}$ matches a planned delta on one row out of eighteen, and MVT − SCHED on these rows ranges from 939 s to 93,535 s, so no threshold isolates them.

We tested a second hypothesis for the final phase. Arrivals in the ranking file keep their labels, and the same defect appears on arrivals. If the defect were a property of the reporting system on a given night, buggy arrivals would point at the nights where departures are at risk. On 2025, no departure defect lies within 24 h of an arrival defect at the same airport (the nearest is 27 h away), the result is identical when the arrivals are shifted by ±7 days as a placebo, and the pure one-day arrival defect occurs twice in 2025 and never in the 2026 ranking file. The defect is per flight, not per night. We therefore make no correction for this family. This is an empirical statement about the fields and signatures we tested, not a proof that identification is impossible.

4 The matching window

The Network Manager flight list carries several off-block clocks. For every departure matched to it in 2025, the airport block time lies within 3606 s of the last filed off-block time:

$$|\text{BLOCK} - \text{LOBT}| \le 3606\,\text{s} \quad \text{on 2,062,251 of 2,062,251 matched departures.} \tag{1}$$

The other clocks satisfy the same bound on 99.997 % (IOBT), 99.97 % (EOBT) and 99.91 % (AOBT) of rows, and the sharp edge of the distribution (Figure 4) identifies the bound as the

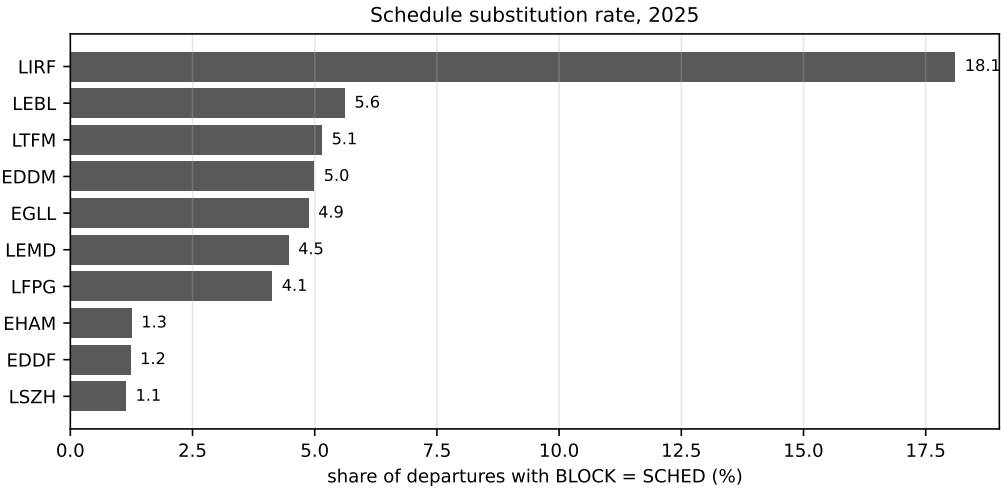


Figure 3: Share of departures whose recorded block time equals the scheduled time to within 10 s, per airport, 2025. On those rows the reported taxi time is the departure delay plus the real taxi.

tolerance of the join between the two sources rather than a property of operations. A movement whose off-block falls outside the window is simply not matched.

Equation (1) has two consequences, and both can only reduce the error.

1. The true taxi time lies in $[\mathrm{MVT} - \mathrm{LOBT} - 3606,\ \mathrm{MVT} - \mathrm{LOBT} + 3606]$. Projecting any prediction on that interval brings it closer to the truth or leaves it unchanged.

2. A schedule substitution writes SCHED into the block field. It is therefore impossible when $|\mathrm{SCHED} - \mathrm{LOBT}| > 3606$ s. In the January and July 2025 holdout, 13,869 departures are in that case and none of them is substituted, yet a classifier that ignores the bound still assigns them a positive substitution probability, weighted by delays that can reach several hours.

Applied after the fact to the holdout predictions of an earlier pipeline (v15), the projection alone reduced the root mean squared error from 345.0 s to 340.7 s, and setting the substitution probability to zero on infeasible rows brought it to 336.7 s. Neither step involves a learnt parameter.

5 Method

5.1 Model

Because the target mixes regimes, a single conditional-mean regressor is badly misspecified: we measured 472.8 s for a single model against 379.1 s for a mixture on an identical feature set in an earlier campaign. The final model (Figure 5) predicts

$$\widehat{y} = \Pi_{\mathrm{LOBT}}\Big[\, p^{\gamma}\,(\mathrm{MVT} - \mathrm{SCHED}) + \big(1 - p^{\gamma}\big)\,\mu \,\Big], \tag{2}$$

where Π_{LOBT} is the projection on the matching window and the components are as follows.

- p is a global XGBoost classifier for schedule substitution, trained only on departures where substitution is feasible, and set to zero elsewhere.

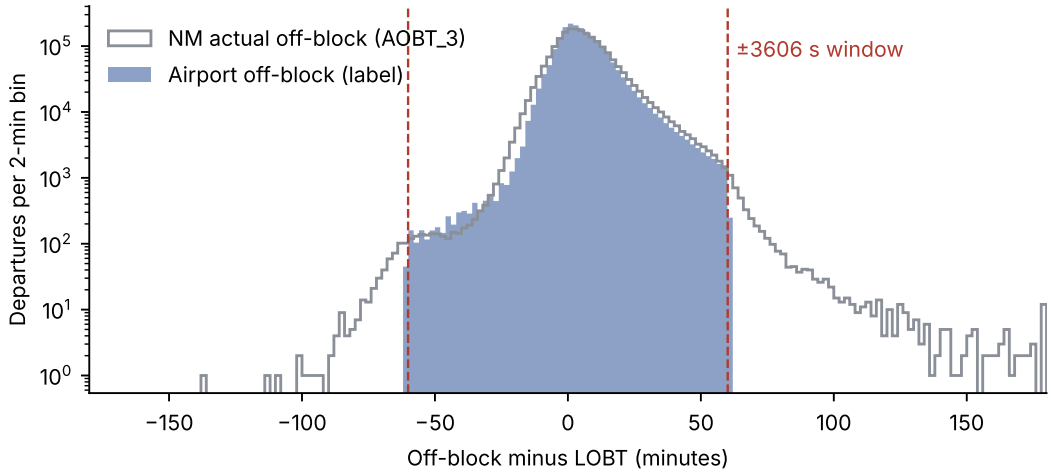


Figure 4: Distance between the off-block and the last filed off-block time, 2025 departures matched to the Network Manager, logarithmic scale. The airport off-block, which defines the label, stops exactly at ±3606 s; the Network Manager actual off-block does not.

- μ is the normal-regime taxi. When the Network Manager off-block gives a plausible taxi, $0 \leq \text{MVT} - \text{AOBT} \leq 7200\,\text{s}$, regressor A predicts the residual over it; otherwise regressor B predicts the taxi directly. Both are trained on non-substituted labels between 30 s and 7200 s, and μ is clipped to that range.
- The exponent γ reflects that a squared loss rewards the expectation over the two regimes, not the most likely regime; it is chosen on the selection fold ($\gamma = 0.9$).
- For unmatched movements without aircraft type and with $\text{MVT} - \text{SCHED} > 3\,\text{h}$, p^{γ} is replaced by the substitution rate of that subset in the training data (Section 3.2).

Three families are trained for regressors A and B: global XGBoost, per-airport XGBoost for regressor A (all ten airports exceed the size threshold of 40,000 rows), and global CatBoost with native categorical handling. Their weights, on a grid of step 0.25, and the exponent are chosen jointly on the selection fold, with an eligibility rule fixed in advance: a blend must beat XGBoost alone by at least 0.5 s, otherwise XGBoost alone is kept. The selected blend is 0.75 CatBoost and 0.25 per-airport XGBoost. The final model is trained on the twelve months of 2025 with 1.1 times the number of trees found by early stopping, three seeds averaged.

5.2 Features

Departure labels (block time and taxi time) are masked in the training data before any contextual feature is built, so that training features are computed under exactly the conditions of the ranking file. The feature set has six groups: plan and clock deltas (takeoff minus each of SCHED, AOBT, LOBT, EOBT and IOBT, the Network Manager delay, the feasibility flag, the day gap); traffic around the takeoff (departures and arrivals within 15 min, same-runway departures 10 min before and 5 min after, gaps to the neighbouring takeoffs); an off-block anchored queue (aircraft taxiing at the flight's off-block, takeoffs between its off-block and its takeoff, recent Network Manager taxi times); the airport climate of the hour and of the day (mean Network Manager taxi and delay, arrival taxi-in, arrival substitution rate); the previous arrival at the same stand; and categorical identifiers (airport, runway, stand, aircraft type, destination, operator, market segment, wake category), plus a stand and runway reference of the Network Manager taxi computed on the training fold. The traffic and climate windows are symmetric around the

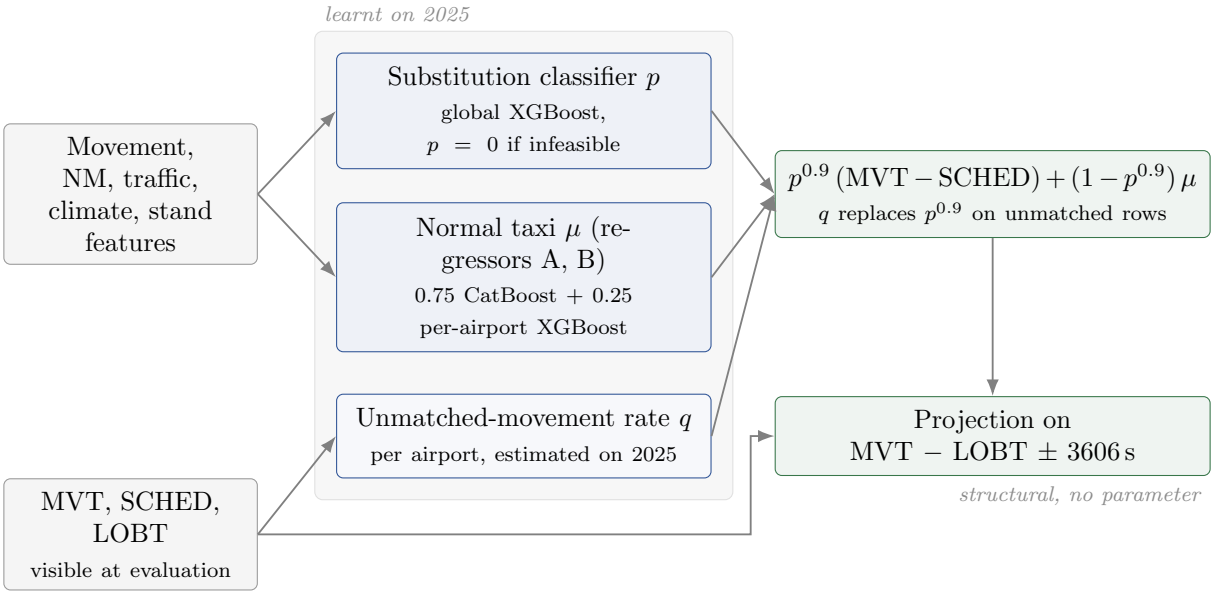


Figure 5: The final model. The substitution classifier, the regressors and the unmatched-movement rate are learnt on 2025; the feasibility condition and the projection follow from the matching window and have no parameter. The blend weights and the exponent are chosen on the selection fold.

takeoff, so they include some later movements, but the dedicated posterior sequence features of our second campaign are excluded; the reason is given in Section 8.

5.3 Validation protocol

Two folds are built on 2025. The selection fold trains on all months except January, April, July and October and validates on April and October; it sets the number of trees by early stopping, the exponent and the blend weights. The control fold trains on all months except January and July, validates on January and July, mimicking the seasonality of the ranking set, and is scored once with the settings of the selection fold. Nothing is chosen on the control fold, and nothing in the final model is chosen with leaderboard feedback.

6 Results

Table 1 gives the selection and control scores of each regressor family and of the selected blend. The blend gained 5.9 s over XGBoost alone on the selection fold; on the control fold it gains 6.7 s overall and 8.9 s on the clean lane, labels up to 20,000 s. Both exceed the 2.9 s range we measured across three seeds of one configuration (Section 9). On this problem CatBoost alone outperforms XGBoost alone, as it outperformed LightGBM in our second campaign, which we attribute to its ordered target statistics on the high-cardinality stand and operator variables.

The gain is spread over all ten airports (Table 2). Against the second-campaign pipeline, the final model is 8.3 s better overall and 3.6 s worse on the clean lane. The clean-lane gap is consistent with the posterior sequence features we left out, worth 3.3 s on the clean lane of the holdout when they were introduced.

Table 3 isolates the structural components. The unmatched rule is worth 6.3 s on the selection fold and 44.3 s on the control fold, because July 2025 carries many unmatched LIRF rows with large delays: it is the component with the largest variance across months, and its contribution to the final score depends on how often the defect occurs in 2026. The projection is worth 1.3 s and 5.3 s.

Table 1: Root mean squared error (seconds) of each regressor family and of the selected blend. The selection fold chose the blend; the control fold is scored once. The clean lane excludes labels above 20,000 s.

	Selection Apr + Oct	Control Jan + Jul	Control, clean lane Jan + Jul
Second-campaign pipeline (v26)		356.60	259.19
XGBoost	221.92	355.00	271.70
Per-airport XGBoost	221.44	354.15	270.59
CatBoost	217.22	348.99	263.65
0.75 CatBoost + 0.25 per-airport	**216.06**	**348.29**	**262.76**

Table 2: Clean-lane root mean squared error (seconds) per airport and month, control fold, XGBoost alone and selected blend.

	January 2025		July 2025	
Airport	XGBoost	Blend	XGBoost	Blend
EDDF	176.9	174.3	198.4	191.0
EDDM	180.7	170.5	187.1	179.8
EGLL	248.0	239.0	307.8	292.0
EHAM	210.1	199.9	175.6	168.2
LEBL	144.8	140.7	268.2	262.5
LEMD	162.9	155.9	191.7	185.9
LFPG	271.3	257.6	260.1	247.7
LIRF	329.2	321.3	710.4	697.8
LSZH	218.9	212.1	205.5	197.6
LTFM	228.4	221.5	290.3	276.3

The final submission was produced on a Kaggle T4 GPU in about 4.8 h, of which CatBoost takes most. The 670,790 identifiers of the submission template all match departures of the final ranking file.

7 How much of the remaining error is reducible?

Table 4 decomposes the squared error of the control fold by regime. The proportions come from a reduced run on 60 % of the training rows (359.9 s overall) and match those of the full run.

Twenty-six rows weigh as much as 323,650 ordinary flights. A single defective record of the rollover family contributes roughly 21,500 to the mean squared error of a two-month validation set; with about twelve such events a year, a two-month evaluation window is a Poisson draw, and the ranking of two models separated by a few seconds is partly decided by it.

On the normal lane, the model regresses towards the mean: taxis below 10 min are over-predicted by about 100 s and taxis of 30 to 60 min are under-predicted by about 400 s. This is the signature of an error in the conditioning variable rather than of a missing regressor. The disagreement between the airport and Network Manager off-blocks has a per-airport standard deviation between 250 s and 500 s; our clean-lane error of 263 s is roughly twice the 75 s to 90 s reported in the literature on cleanly measured taxiing. Under a squared loss, shrinkage towards the mean is the optimal response to such noise, and a different model fed the same columns cannot undo it. The information that could, a physical observation of the push-back, is not in the provided data: surface ADS-B tracks are the obvious candidate, and we did not use them.

For the unmatched movements, a perfect substitution classifier on that lane would remove

Table 3: Ablation of the structural components of the selected model, root mean squared error in seconds.

Model	Selection	Control
Full model	216.06	348.29
Without the unmatched-movement rule	222.40	392.55
Without the rule and the projection	223.71	397.87

Table 4: Decomposition of the control-fold mean squared error by regime.

Regime	Rows	Share of MSE	Reducible with the provided columns?
Labels above 20,000 s	26	40 %	No identifying signal found
Normal departures	323,650	40 %	Measurement noise
Unmatched to the Network Manager	5227	15 %	At most 10 points, with a perfect classifier
Schedule substitution	15,324	5 %	Classifier plateaus at AUC 0.88

10 of its 15 points of squared error, an upper bound that no realistic classifier approaches. We conclude that the large remaining gains lie outside the provided columns, and we do not claim that the remaining gap to the leading models is irreducible with external data.

8 Directionality of the congestion screening

The strongest predictor in the taxi-time literature is the state of the departure queue at pushback [1, 2]. We implemented it twice, in two independent forms: queue length and recent throughput anchored on the aircraft's own off-block, and a demand queue counting flights whose planned pushback preceded the takeoff without having departed. Both were neutral to slightly negative. The explanation is conditioning: once the Network Manager off-block is observed, the model already knows the duration of the flight's own ground phase as measured by the second source, and the queue that produced that duration adds nothing.

The screening is directional, however. The ranking file contains every movement of the evaluation months, including the departures that took off after the target flight. Features built from those (intervals to the next takeoffs on the same runway, the position of the flight within a burst of closely spaced takeoffs, and above all the number of the next ten departures that had already pushed back when the target took off) gained 3.62 s on the leaderboard in our second campaign and improved eight airports out of ten on the holdout. That last feature is the queue behind the aircraft, and it is measurable only by looking forward.

These features have no operational meaning. A controller does not know future takeoffs, and a model using them cannot be deployed for departure metering. They are admissible in a retrospective reconstruction task, which is what the challenge is, but they change what a reported score means. For the final model we left out these dedicated forward-looking features, keeping only symmetric traffic and climate windows around the takeoff, at an estimated cost of about 3 s on the clean lane. We believe the distinction between forecasting and reconstruction is worth making explicit in challenge design, since it silently changes the admissible feature space.

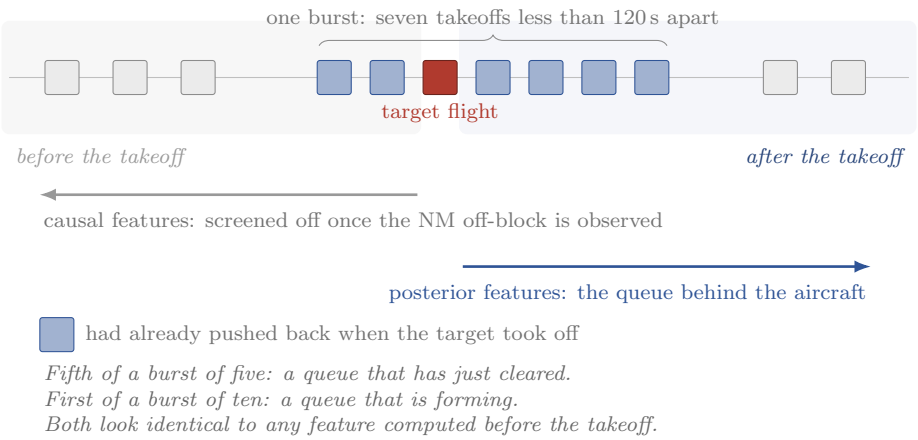


Figure 6: Why the congestion screening is directional. Each square is a takeoff on the same runway. Features computed from earlier takeoffs add nothing once the Network Manager off-block is observed, because that timestamp already encodes the queue the aircraft went through. Features computed from later takeoffs, in particular the count of subsequent departures that had already left their stand, measure the queue the aircraft was part of, and no causal feature can substitute for them.

9 Negative results and measured variability

9.1 What did not work

Table 5 lists the substantive ideas we tested and rejected across the three campaigns. We report them because several are ideas a reader would otherwise be likely to try.

The recalibration deserves a note. EUROCONTROL publishes, per airport and month, the number of departures and the total reference and additional taxi-out time, from which a mean taxi-out follows, including for the months of the final file. In spring and summer this mean agrees with the mean of our clean labels within about 15 s at eight airports out of ten; in winter it lies 50 s to 166 s below them at EDDF, EDDM, LFPG and LSZH, a gap consistent with de-icing being counted differently. Our model's airport-month means were already within 20 s of the truth on the control fold, so shifting them towards the published series made predictions worse. A published aggregate is only a useful anchor if its definition matches the label.

The rollover correction teaches a related lesson. It was validated by a month-transfer protocol within 2025, improved both validation months, and still lost on the leaderboard: with about twelve events a year, a two-month window does not estimate a rate. Modelling a data defect from its historical frequency assumes the producing system is statistically stationary at the evaluation scale, an assumption that is not testable within the data.

9.2 How large is a real effect?

In our second campaign, three seeds of a single fixed configuration gave 366.43 s, 366.17 s and 363.52 s, a range of 2.9 s. Three runs do not estimate a variance, and random training and data partitioning are known to produce non-negligible performance variation in model comparisons [4]; we use the range as a warning threshold. It renders several of our own earlier single-seed comparisons unresolved. The components of the final model clear it with margin: the blend gains 6.7 s on a fold where it was not selected, and the structural components of Section 4 involve no learnt parameter.

Table 5: Rejected hypotheses. Effects are on 2025 validation data unless stated otherwise.

Hypothesis	Outcome
Recalibrate airport-month means on EUROCONTROL's published monthly taxi-out times	Model means already within 20 s of the truth; the series differs from the labels by up to 166 s in winter. Costs 0.7 s to 3.1 s out of time
Flag rollover defects through arrival defects on the same night	No co-occurrence; placebo at ± 7 days identical
Expectation correction for the rollover family, calibrated on 2025 rates	19.8 s on holdout, -2.45 s on the leaderboard
Single regressor instead of the regime mixture	Costs 94 s
Queue and throughput features anchored on pushback	Neutral, twice
Demand queue anchored on scheduled and estimated off-block	-1.6 s
Degraded-day detection from arrival taxi-in	No effect: a de-icing aircraft delays itself, not the arrivals
Meteorological features (temperature, dew-point spread, precipitation codes)	Degraded July: temperature acts as a season surrogate
Upper clipping of predictions	Degrades at every threshold: the extreme labels are real
Open ADS-B derived off-block times (OPDI)	Ground coverage between 0.2 % and 16 %; first detection trails the off-block by 700 s to 1300 s

10 Earlier campaigns

The final model is the third of three campaigns. The first built the regime mixture and selected its exponent on the leaderboard over the grid $\{0.5, 0.6, 0.75, 1.0\}$; the public and holdout curves were parallel with an offset of about 32 s (Figure 7), our first evidence that holdout gains transfer for physical properties of the model. The second campaign added the reconstruction rule, the posterior sequence features, per-airport regressors with a global classifier and a CatBoost ensemble; Table 6 traces its public progression. Because several of those settings were chosen or confirmed with repeated leaderboard queries, which can overfit a public leaderboard [3], we treat these gains as descriptive. The third campaign kept the ideas that rest on structure or on out-of-time validation and re-selected every setting on 2025.

11 Discussion

11.1 Implications for operational data

Two major terms in our residual are produced by the reporting chain: the disagreement between two off-block definitions, and defects whose provenance is hidden. For the data producers, publishing a provenance flag alongside the block time, a substitution indicator where the scheduled time was used as a fallback, and a boolean recording that a date was inferred rather than observed would be worth more than any modelling applied downstream. The matching window of Section 4 illustrates the converse: a single documented property of the join removed more error than most of the modelling changes we tried.

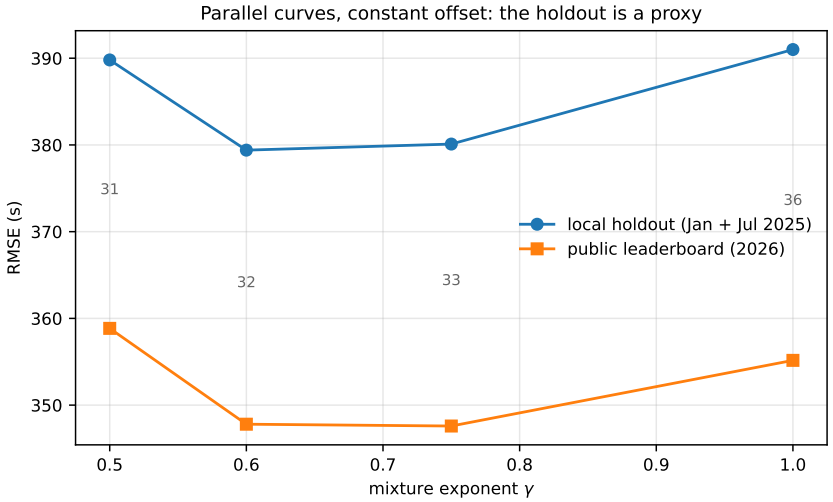


Figure 7: Holdout and public root mean squared error as a function of the mixture exponent γ, first campaign. The two curves are parallel with an offset of about 32 s.

Table 6: Public leaderboard progression of the second-campaign pipeline. Each row adds one change to the row above.

Submission	Change	Public RMSE (s)
v19	Merged baseline pipeline	338.37
v21	Label reconstruction rule (Section 3.2)	330.42
v22	Posterior sequence features (Section 8)	326.80
v23	Double capacity, lean classifier, three-seed average	325.15
v25	Per-airport regressors, global classifier, CatBoost	320.10
v26	CatBoost weight 0.75	**319.21**

11.2 Limitations

CatBoost reached its iteration cap in the selection fold (6000 trees for regressor A, 4000 for regressor B) without triggering early stopping, so the final blend is likely slightly under-trained. February and June are in the final file but not in a validation fold; they exist in the 2025 training data and lie seasonally between the two folds. The unmatched-movement rule depends on the frequency of an airport-side defect and its contribution varies widely between months. The error decomposition of Table 4 comes from a reduced run. Our evaluation rests on months of 2025 as a proxy for months of 2026, which cannot detect non-stationarity of the kind that defeated the rollover correction, and we had no access to the systems that produce the block times, so our characterisation of the defects is inferred from their traces.

11.3 Reproducibility

The repository provides the final pipeline as a single script (`src/taxiout_general.py`) and as the Kaggle notebook that produced the submission, with fixed seeds. A run writes the selected settings, the fold scores and the out-of-fold predictions of both folds. The full run takes about 4.8 h on a Kaggle T4 GPU, and 17 min with XGBoost alone. The final model uses only challenge data. The diagnostics behind the counts of Sections 3 and 4, the co-occurrence test, the complete experiment log and the earlier pipelines are kept in the repository; the challenge data are not redistributed.

12 Conclusion

The taxi-out times distributed in this challenge are, for a small but decisive fraction of departures, not taxi-out times. We characterised two families of defects, recovered one from visible columns and found no identifier for the other. We showed that the airport off-block never leaves a one-hour window around the last filed off-block, a property of the join that bounds every prediction and rules out substitution on a known subset of flights at no modelling cost. On that basis we built a generalist model selected without leaderboard feedback, which outperforms our earlier leaderboard-tuned pipeline on the same holdout and whose selected components transfer from one validation fold to another. Most of the error that remains lies in a few unidentifiable records and in the disagreement between two off-block definitions; reducing it further calls for new information, such as a physical observation of the push-back, rather than a different model of the same columns.

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References

[1] Idris, H., Clarke, J.-P., Bhuva, R., and Kang, L., "Queuing Model for Taxi-Out Time Estimation," *Air Traffic Control Quarterly*, Vol. 10, No. 1, 2002.

[2] Ravizza, S., Chen, J., Atkin, J. A. D., Burke, E. K., and Stewart, P., "Aircraft Taxi Time Prediction: Comparisons and Insights," *Applied Soft Computing*, Vol. 14, 2014, pp. 397–406, doi:10.1016/j.asoc.2013.10.004.

[3] Blum, A., and Hardt, M., "The Ladder: A Reliable Leaderboard for Machine Learning Competitions," *Proceedings of the 32nd International Conference on Machine Learning*, Vol. 37, 2015, pp. 1006–1014.

[4] Raste, S., Singh, R., Vaughan, J., and Nair, V. N., "Quantifying Inherent Randomness in Machine Learning Algorithms," arXiv:2206.12353, 2022, doi:10.48550/arXiv.2206.12353.